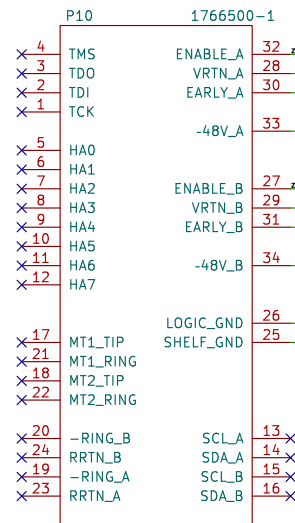
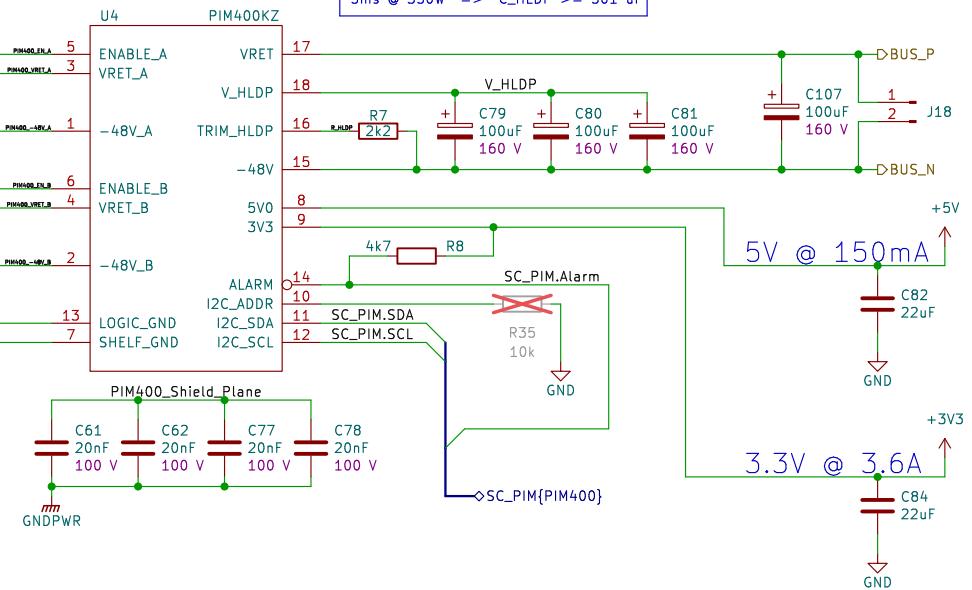


Zone 1 Connector (P10)



-48V Power

V_HLDP = 91 V => R_TRIM = 2.2 k
 5ms @ 400W => C_HLDP >= 572 uF
 5ms @ 350W => C_HLDP >= 501 uF



I2C Address = 8'b0101_111X, 8'h5E, 7'h2F

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Sheet: /ATCA_Z1/

File: ATCA_Z1.kicad_sch

Title: ATCA Simple Loopback HUB

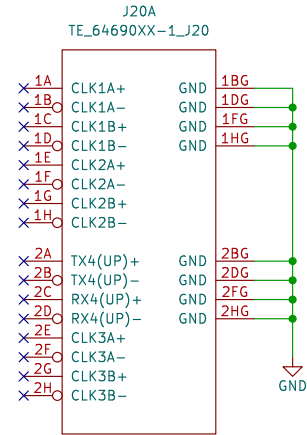
Size: A4 Date: 2024-09-18

KiCad E.D.A. 8.0.8

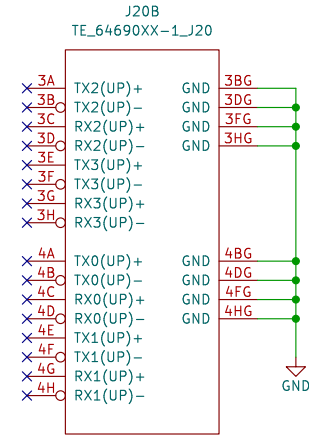
Rev: 1.0

Id: 2/14

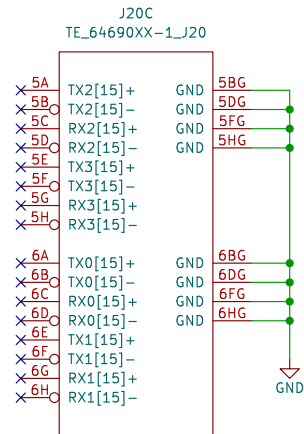
CLOCK INTERFACE



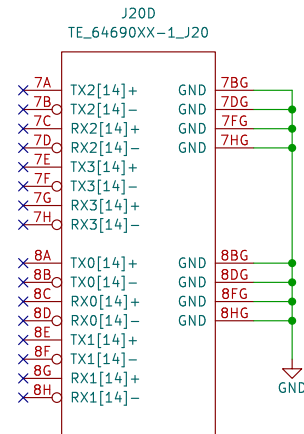
UPDATE INTERFACE



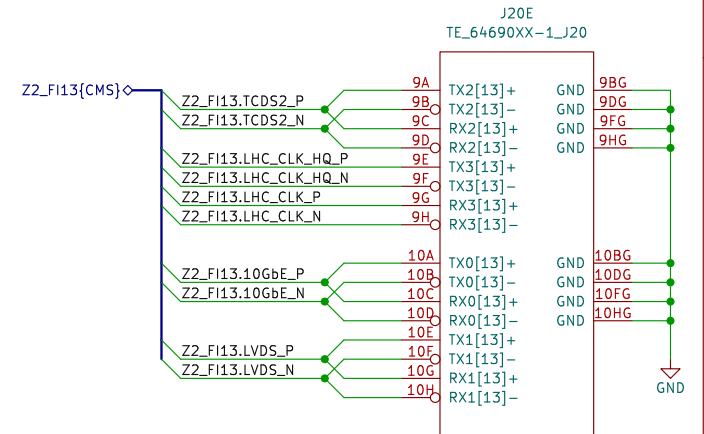
FABRIC INTERFACE 15



FABRIC INTERFACE 14



FABRIC INTERFACE 13



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Sheet: /J20/

File: ATCA_Z2_J20.kicad_sch

Title: ATCA Simple Loopback HUB

Size: A4

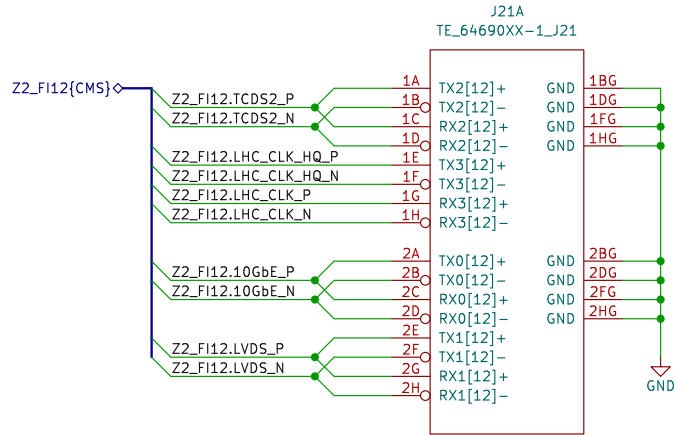
Date: 2024-09-18

Rev: 1.0

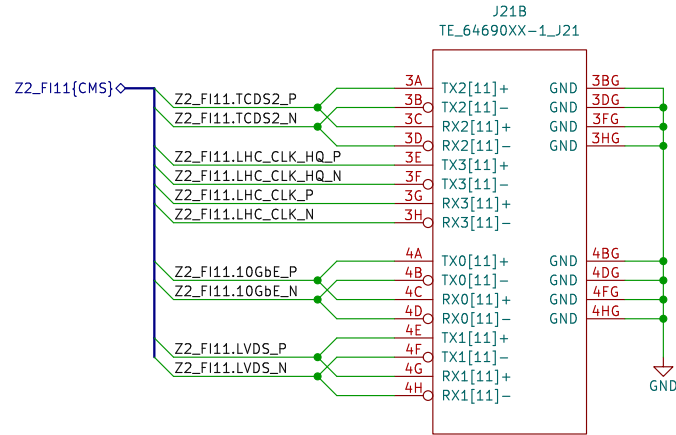
KiCad E.D.A. 8.0.8

Id: 3/14

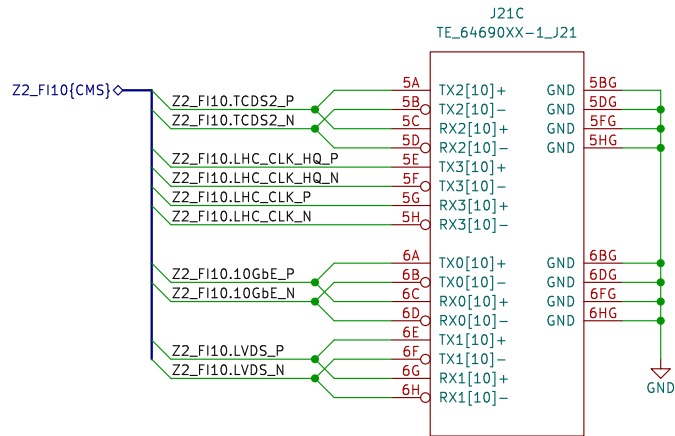
FABRIC INTERFACE 12



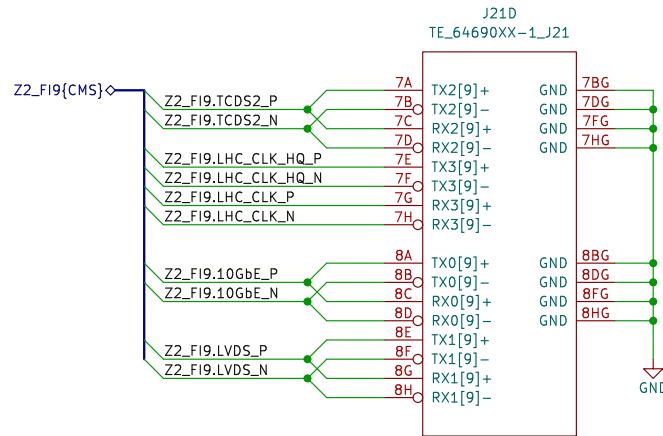
FABRIC INTERFACE 11



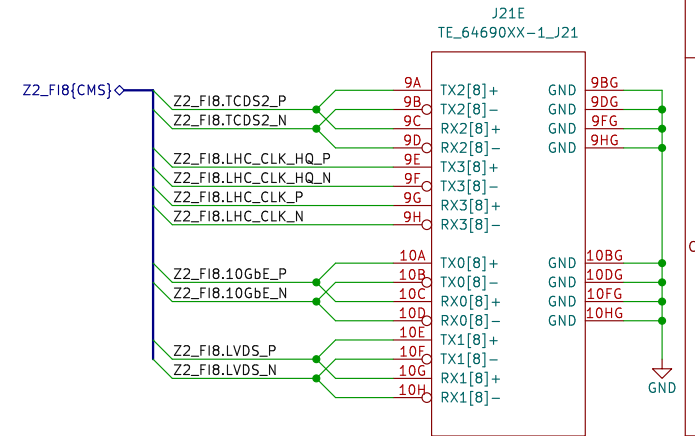
FABRIC INTERFACE 10



FABRIC INTERFACE 9



FABRIC INTERFACE 8



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Sheet: /J21/

File: ATCA_Z2_J21.kicad_sch

Title: ATCA Simple Loopback HUB

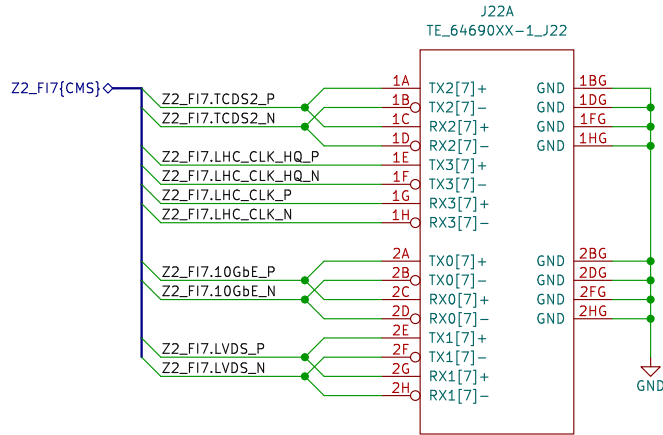
Size: A4 Date: 2024-09-18

KiCad E.D.A. 8.0.8

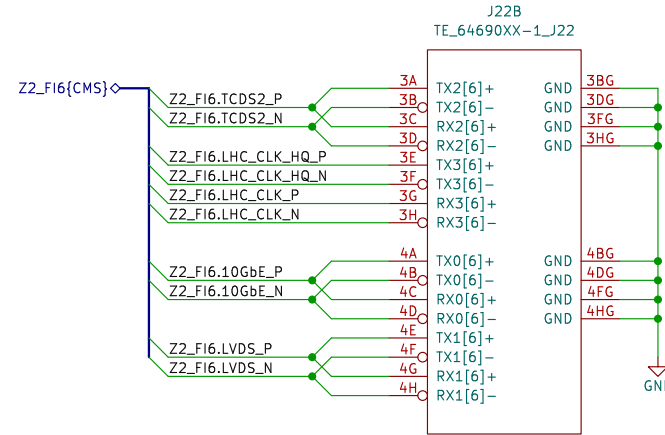
Rev: 1.0

Id: 4/14

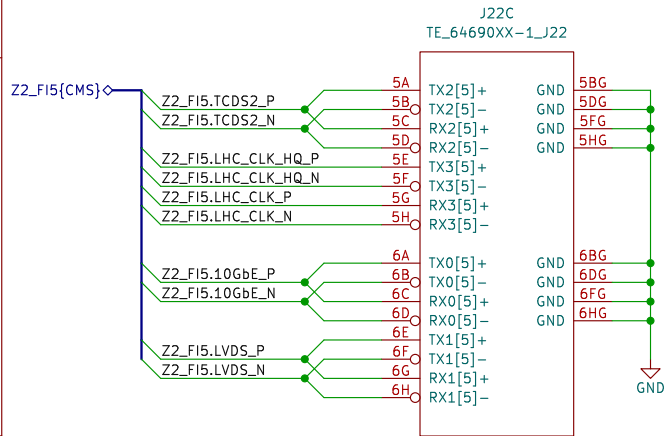
FABRIC INTERFACE 7



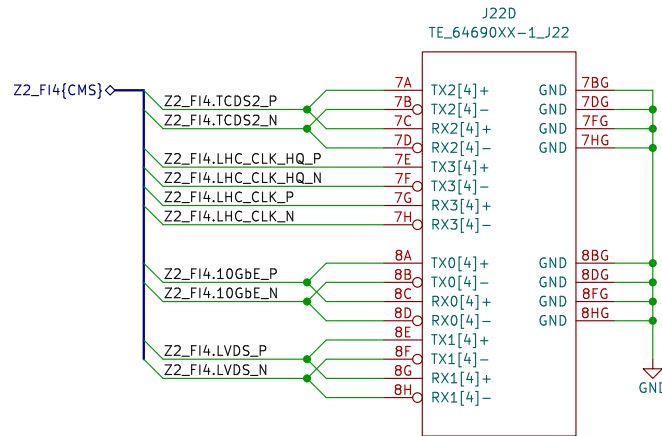
FABRIC INTERFACE 6



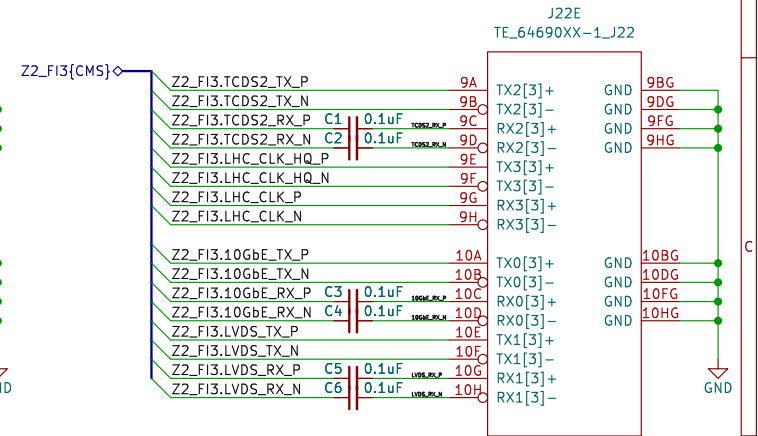
FABRIC INTERFACE 5



FABRIC INTERFACE 4



FABRIC INTERFACE 3



- NOTES:
- 1) Fabric Interface (FI) 3 is the only one that is broken out to the front panel
 - 2) FI3 has capacitors at the RX channel since the TX channel has capacitors at the receiving RX channel in the slot board

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Sheet: /J22/
File: ATCA_Z2_J22.kicad_sch

Title: ATCA Simple Loopback HUB

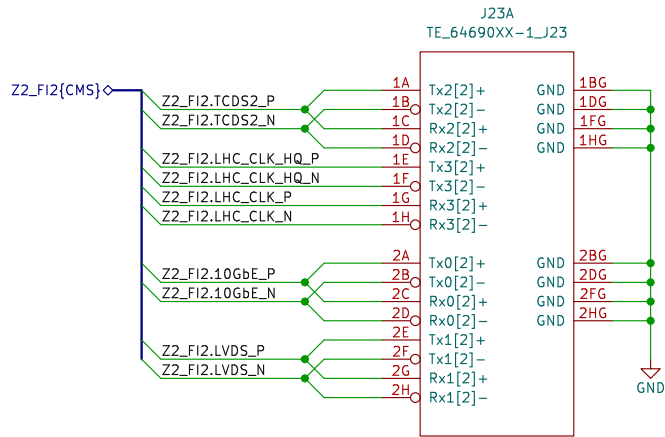
Size: A4 Date: 2024-09-18

KiCad E.D.A. 8.0.8

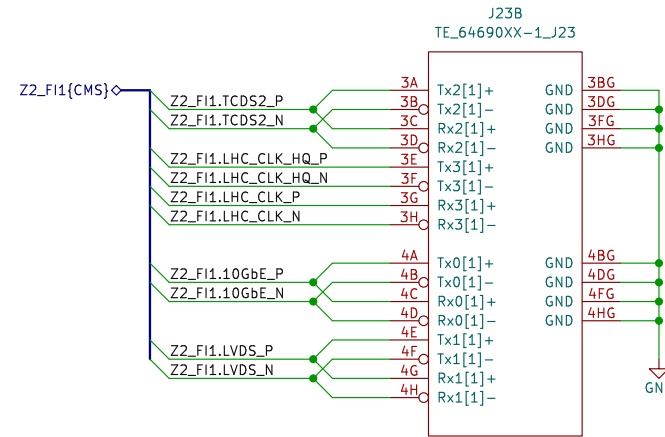
Rev: 1.0

Id: 5/14

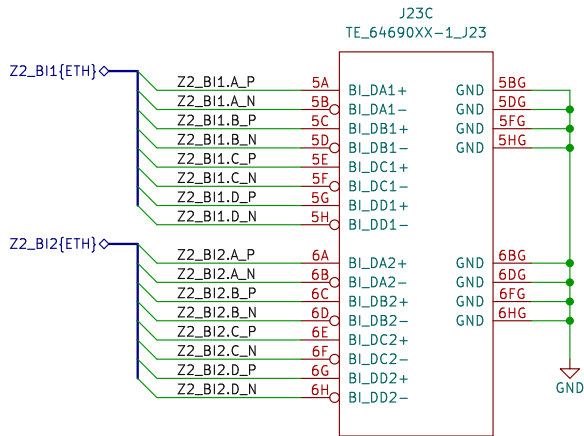
FABRIC INTERFACE 2



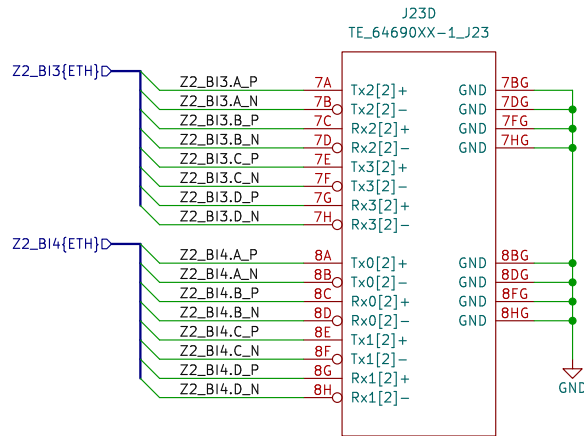
FABRIC INTERFACE 1



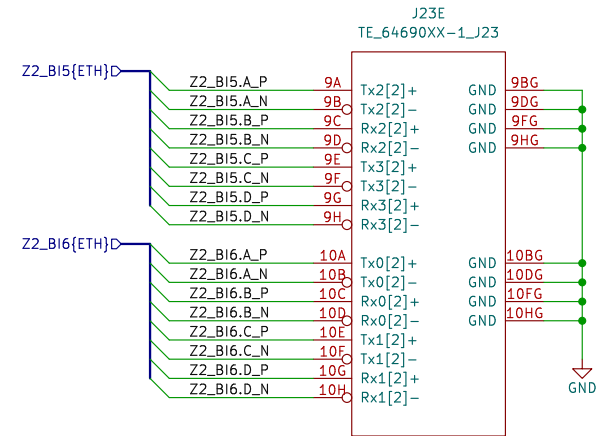
BASE INTERFACE 1-2



BASE INTERFACE 3-4



BASE INTERFACE 5-6



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Sheet: /J23/

File: ATCA_Z2_J23.kicad_sch

Title: ATCA Simple Loopback HUB

Size: A4

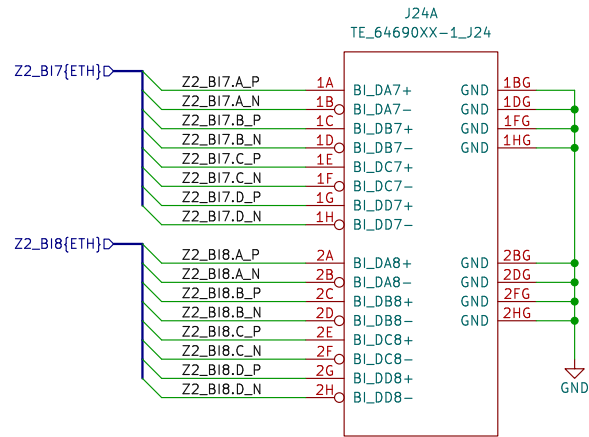
Date: 2024-09-18

KiCad E.D.A. 8.0.8

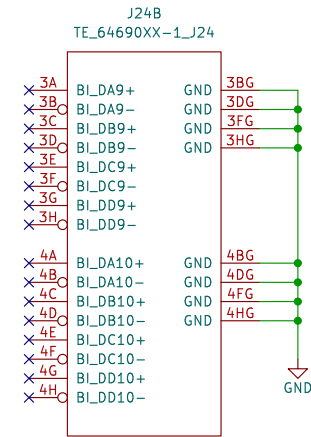
Rev: 1.0

Id: 6/14

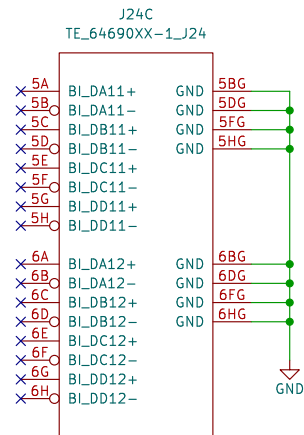
BASE INTERFACE 7–8



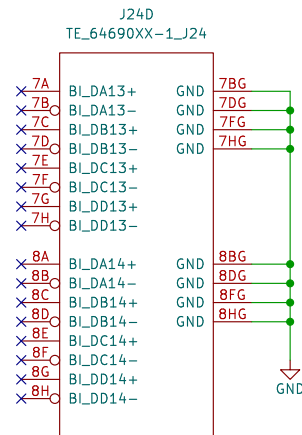
BASE INTERFACE 9–10



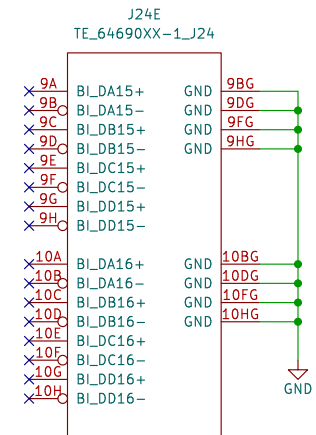
BASE INTERFACE 11–12



BASE INTERFACE 13–14



BASE INTERFACE 15–16



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Sheet: /J24/

File: ATCA_Z2_J24.kicad_sch

Title: ATCA Simple Loopback HUB

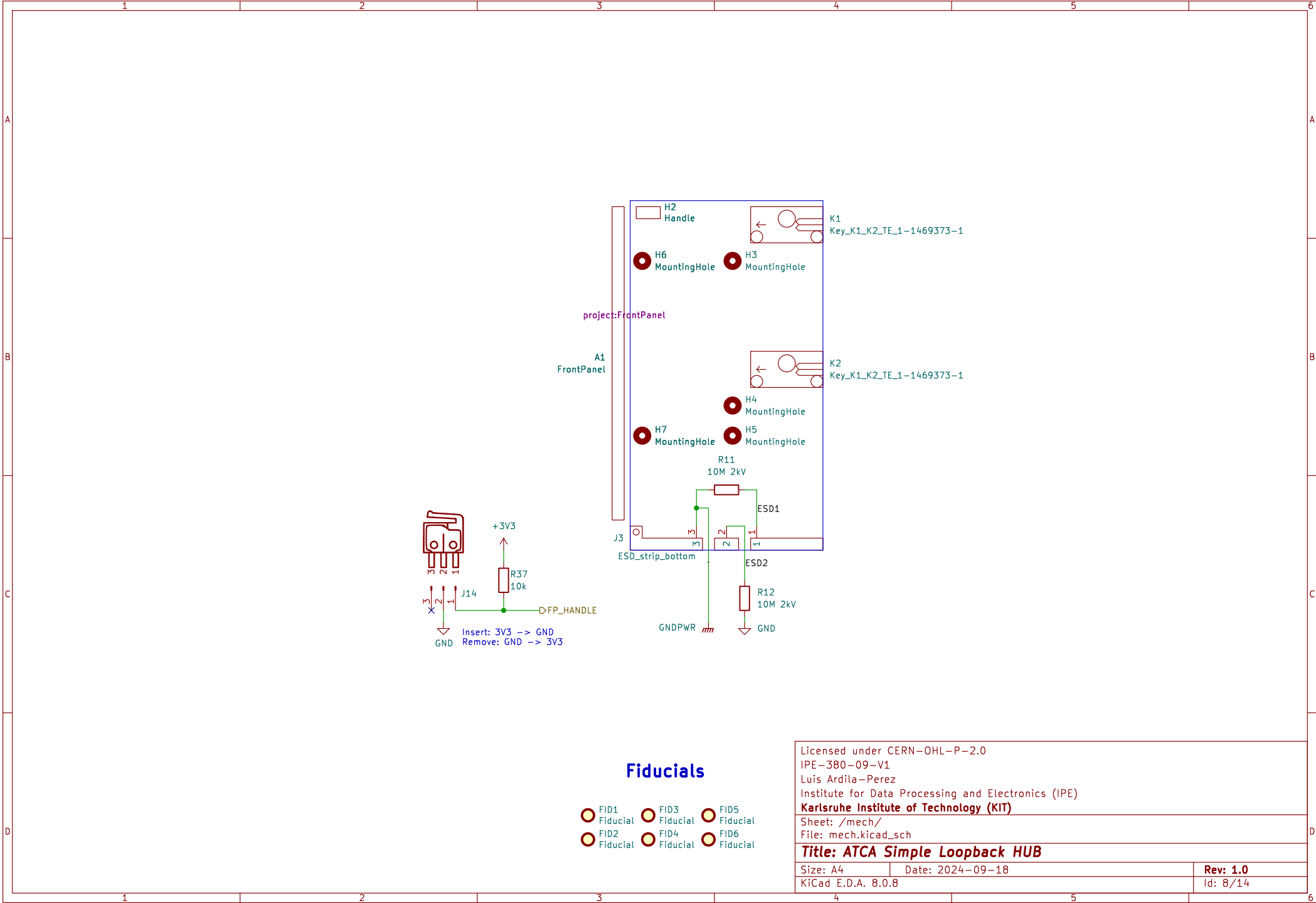
Size: A4

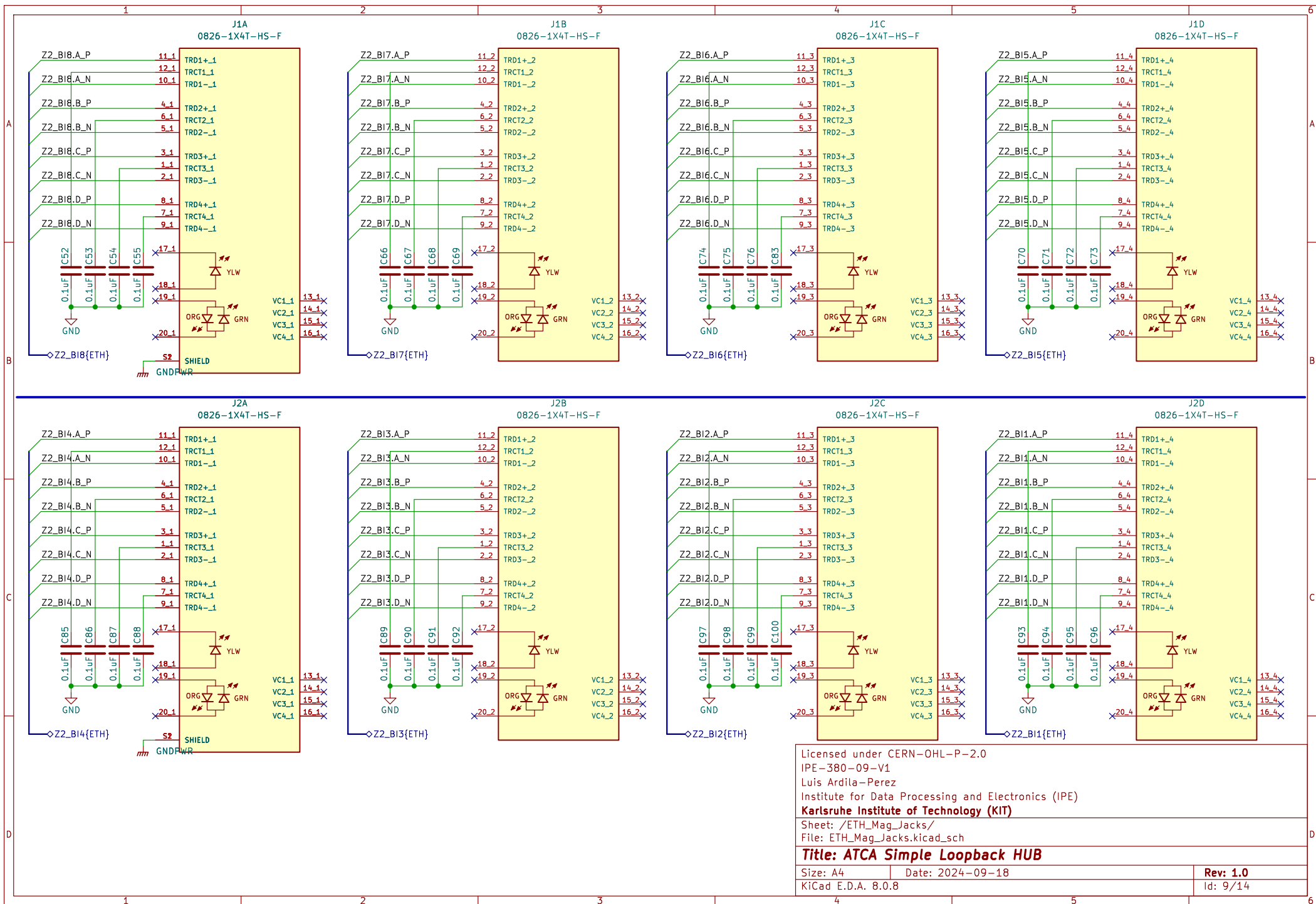
Date: 2024-09-18

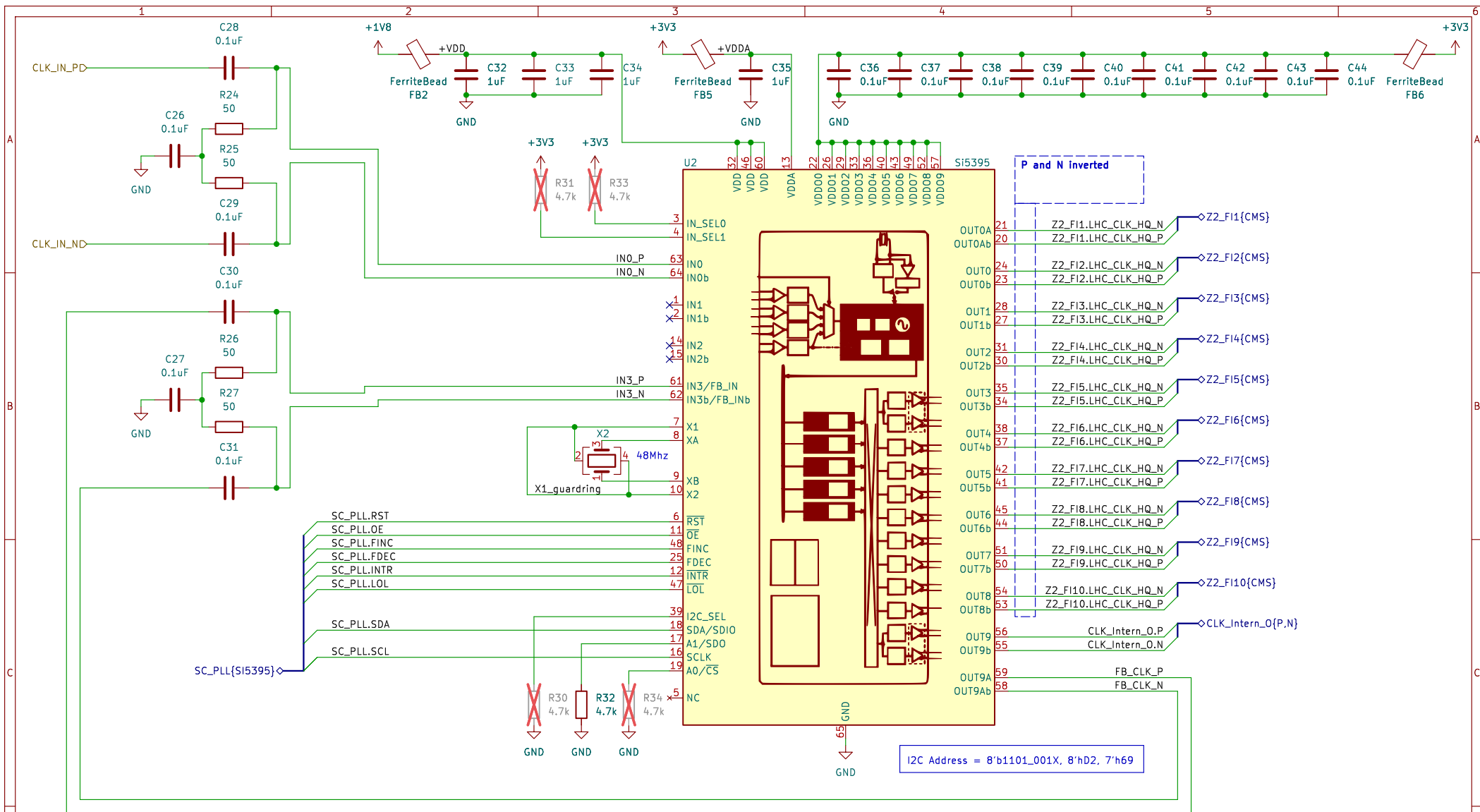
Rev: 1.0

KiCad E.D.A. 8.0.8

Id: 7/14







NOTES:

- 1) IN_SELL (I): Input Reference Select, pins are internally pulled low
- 2) RSTb (I): Device Reset. This pin is internally pulled-up and can be left unconnected when not in use
- 3) OEB (I): Output Enable. This pin disables all outputs when held high. This pin is internally pulled low and can be left unconnected when not in use.
- 4) FINC (I): Step-up the output frequency of a selected output. This pin is internally pulled low and can be left unconnected when not in use.
- 5) FDEC (I): Step-down the output frequency of a selected output. This pin is internally pulled low and can be left unconnected when not in use.
- 6) INTRb (O): Interrupt, asserted low when a change in device status has occurred. It should be left unconnected when not in use.
- 7) LOLb (O): Loss of Lock, indicates when the DSPLL is locked (high) or out-of-lock (low). It can be left unconnected when not in use
- 8) I2C_SEL (I): I2C (I2C_SEL = 1) or SPI (I2C_SEL = 0). This pin is internally pulled up by a ~ 20 kΩ resistor to the voltage selected by the IO_VDD_SEL reg
- 9) A1 (I): In I2C mode, this pin is open-drain and functions as the A1 address input pin. It does not have an internal pull-up or pull-down resistor.
- 10) A0 (I): hardware controlled address A0 in I2C mode. This pin is internally pulled-up by a ~20 kΩ resistor and can be left unconnected when not in use.
- 11) Power: 1V8 @ 260mA & 3V3 @ 130 + 30 * 12 = 490mA

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Sheet: /LHC_CLK_HQ/

File: LHC_CLK_HQ.kicad_sch

Title: ATCA Simple Loopback HUB

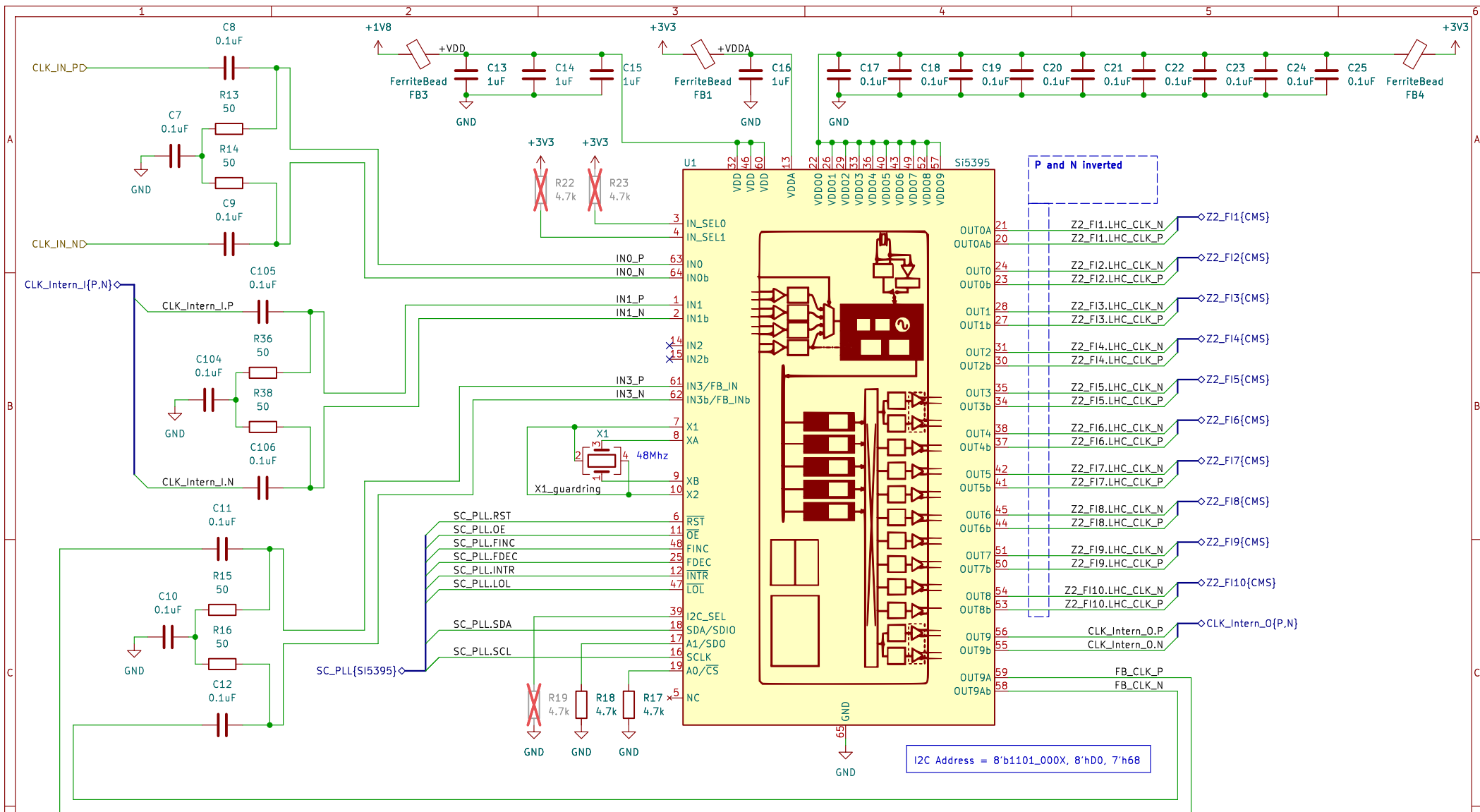
Size: A4

Date: 2024-09-18

KiCad E.D.A. 8.0.8

Rev: 1.0

Id: 10/14



NOTES:

- 1) IN_SELL (I): Input Reference Select, pins are internally pulled low
- 2) RSTb (I): Device Reset. This pin is internally pulled-up and can be left unconnected when not in use.
- 3) OEB (I): Output Enable. This pin disables all outputs when held high. This pin is internally pulled low and can be left unconnected when not in use.
- 4) FINC (I): Step-up the output frequency of a selected output. This pin is internally pulled low and can be left unconnected when not in use.
- 5) FDEC (I): Step-down the output frequency of a selected output. This pin is internally pulled low and can be left unconnected when not in use.
- 6) INTRb (O): Interrupt, asserted low when a change in device status has occurred. It should be left unconnected when not in use.
- 7) LOLb (O): Loss of Lock, indicates when the DSPLL is locked (high) or out-of-lock (low). It can be left unconnected when not in use.
- 8) I2C_SEL (I): I2C (I2C_SEL = 1) or SPI (I2C_SEL = 0). This pin is internally pulled up by a ~ 20 kΩ resistor to the voltage selected by the IO_VDD_SEL reg
- 9) A1 (I): In I2C mode, this pin is open-drain and functions as the A1 address input pin. It does not have an internal pull-up or pull-down resistor.
- 10) A0 (I): hardware controlled address A0 in I2C mode. This pin is internally pulled-up by a ~20 kΩ resistor and can be left unconnected when not in use.
- 11) Power: 1V8 @ 260mA & 3V3 @ 130 + 30 * 12 = 490mA

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Sheet: /LHC_CLK/

File: LHC_CLK.kicad_sch

Title: ATCA Simple Loopback HUB

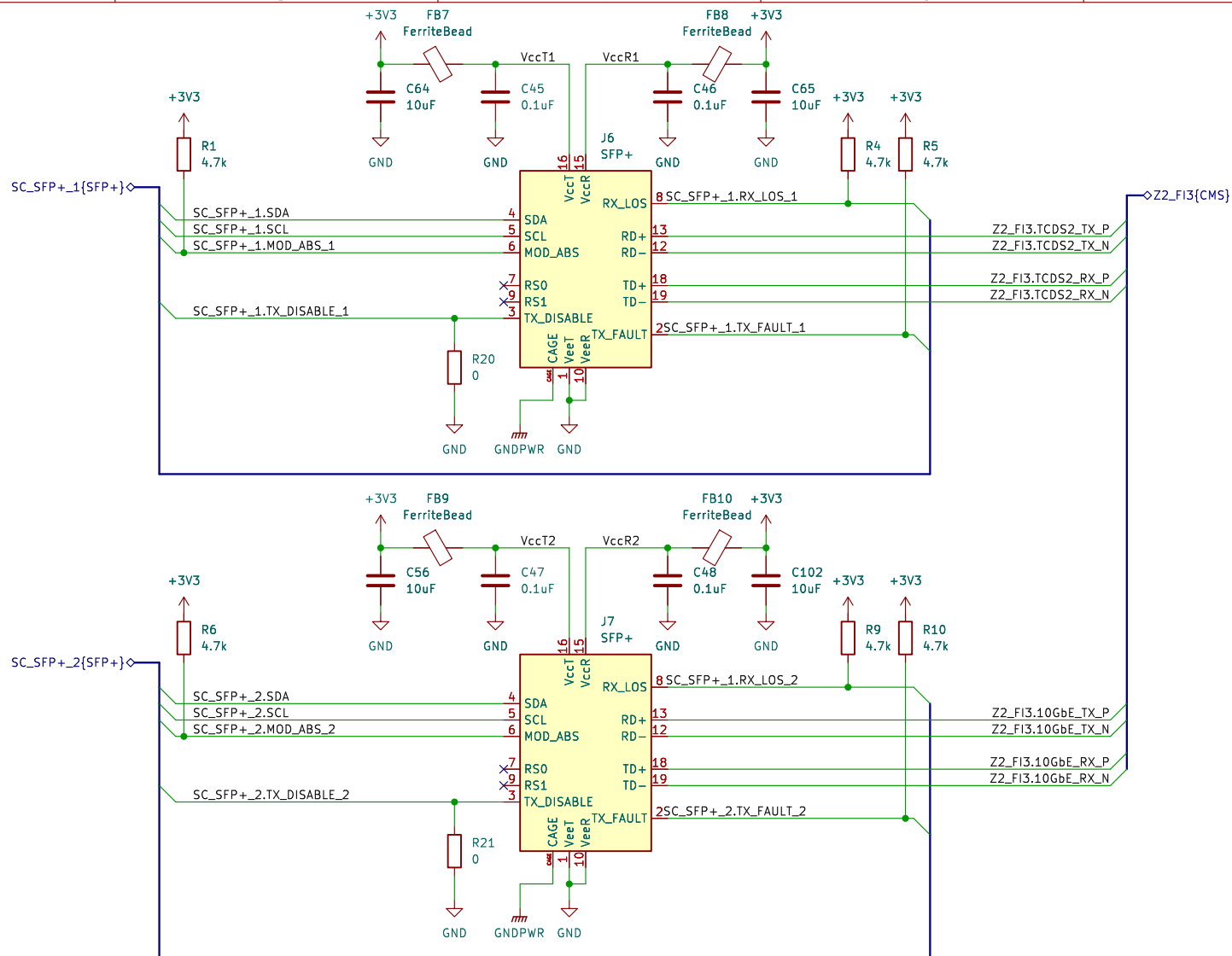
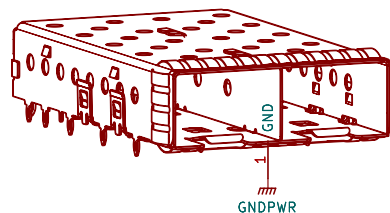
Size: A4

Date: 2024-09-18

KiCad E.D.A. 8.0.8

Rev: 1.0

Id: 11/14



NOTES:

- 1) TX_Fault: TX_Fault is a module output pin that when High, indicates that the module transmitter has detected a fault condition related to laser operation or safety. The TX_Fault output pin is an open drain/collector and must be pulled up to the Host_Vcc with 4.7k–10k ohms on the host board.
- 2) TX_Disable: TX_Disable is a module input pin. When TX_Disable is asserted High or Left open, the SFP+ module transmitter output must be turned off. The TX_DIS pin must be pulled up to VccT in the SFP+ module with 4.7k to 10k resistor.
- 3) RS0/RS1: RS0 and RS1 are module input rate select pins and are pulled low to VeeT with a >30kΩ resistor in the module. RS0 is an input hardware pin which optionally selects the optical receive data path rate coverage for an SFP+ module. RS1 is an input hardware pin which optionally selects the optical transmit path data rate coverage for an SFP+ module.
- 4) MOD_ABS: Mod_ABS is pulled up to Host_Vcc with 4.7kΩ–10kΩ on the host board and connected to VeeT or VeeR in the SFP+ module. MOD_ABS is then asserted “High” when the SFP+ module is physically absent from a host slot. In the SFP MSA (INF8074) this pin had the same function but is called MOD_DEFO.
- 5) SCL/SDA: SCL is the 2-wire interface clock and SDA is the 2-wire interface data line. SCL and SDA are pulled up to a voltage in the range of 3.14V to 3.46V on the host.
- 6) RX_LOS: RX_LOS when High indicated an optical signal level below that specified in the relevant standard. The RX_LOS pin is an open drain/collector output and must be pulled up to host Vcc with a 4.7kΩ–10kΩ on the host board. RX_LOS assert min and de-assert max are defined in the relevant standard.
- 7) Power can be up to 2.5W each SFP+

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Sheet: /SFP+/

File: SFP+.kicad_sch

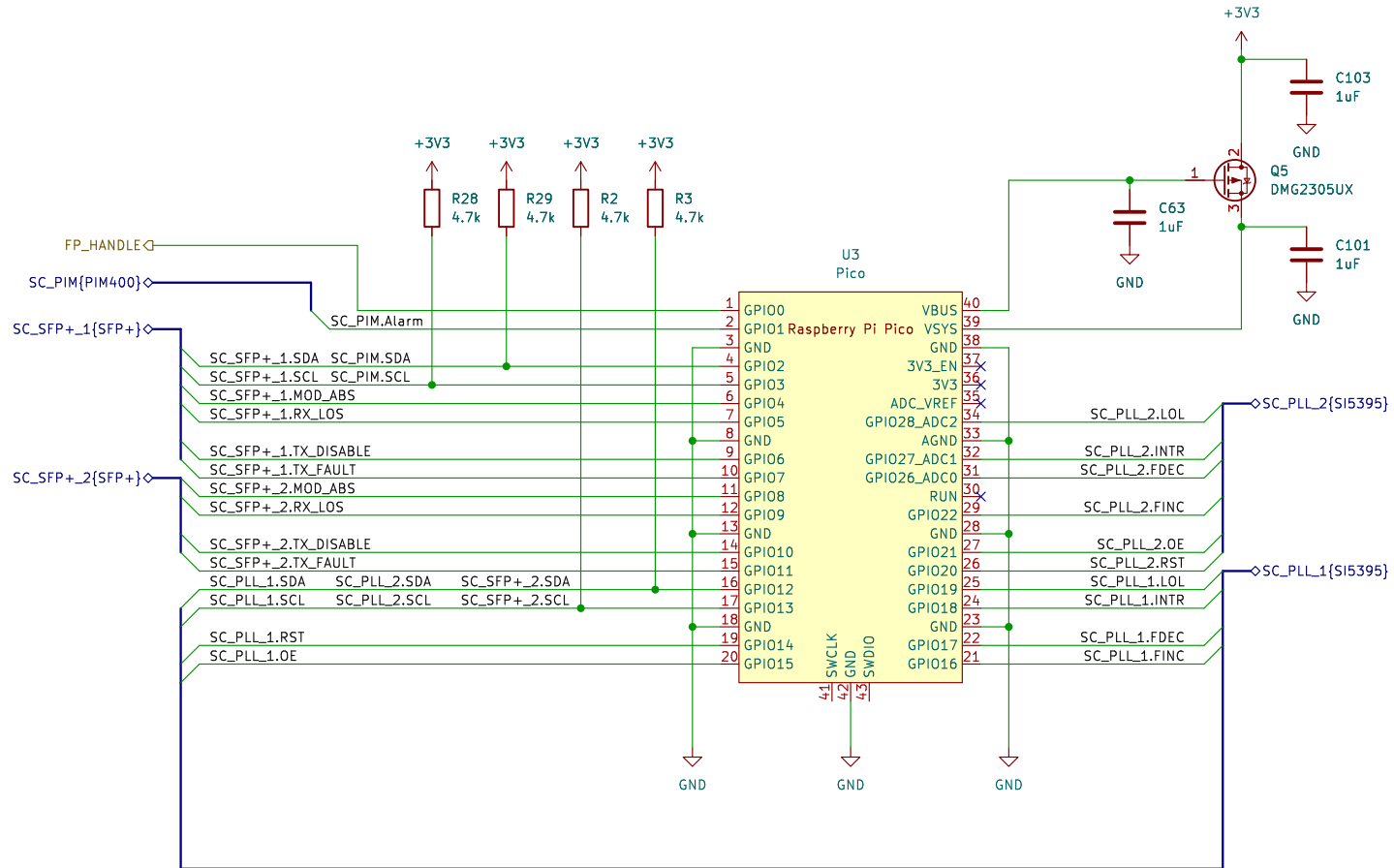
Title: ATCA Simple Loopback HUB

Size: A4 Date: 2024–09–18

KiCad E.D.A. 8.0.8

Rev: 1.0

Id: 12/14



I2C BUS 0

SFP+_2
I2C Address = 8'b1010_000X, 8'hA0, 7'h50

SI5395_1
I2C Address = 8'b1101_000X, 8'hD0, 7'h68

SI5395_2
I2C Address = 8'b1101_001X, 8'hD2, 7'h69

I2C BUS 1

SFP+_1
I2C Address = 8'b1010_000X, 8'hA0, 7'h50

PIM400
I2C Address = 8'b0101_111X, 8'h5E, 7'h2F

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Sheet: /rpi-pico/
File: rpi-pico.kicad_sch

Title: ATCA Simple Loopback HUB

Size: A4 Date: 2024-09-18

KiCad E.D.A. 8.0.8

Rev: 1.0

Id: 13/14

